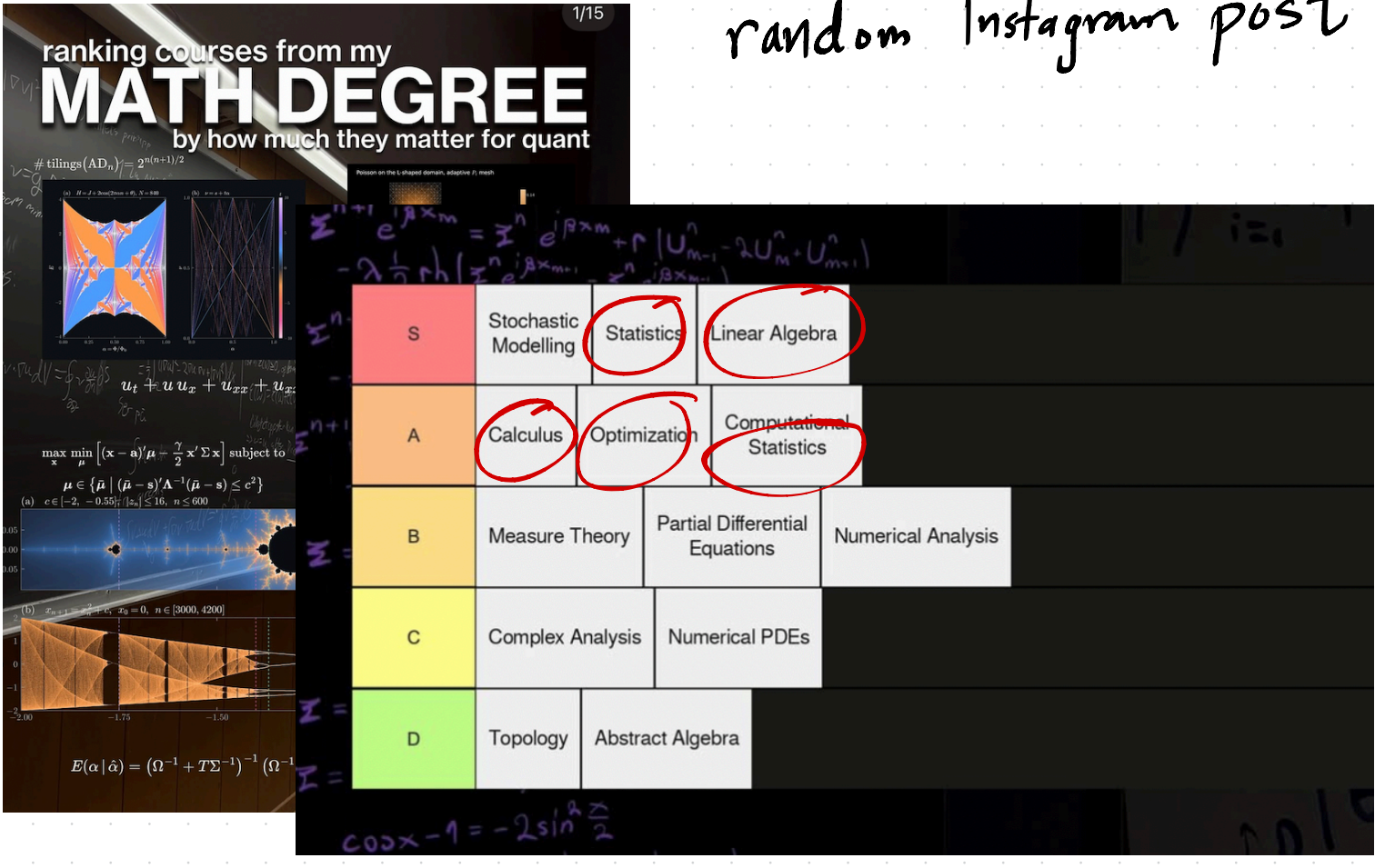
EECS 245, Fall 2026

LEC 18 Gradients and Gradient Descent

→ Read: Ch. 8.2 - 8.3

random Instagram post



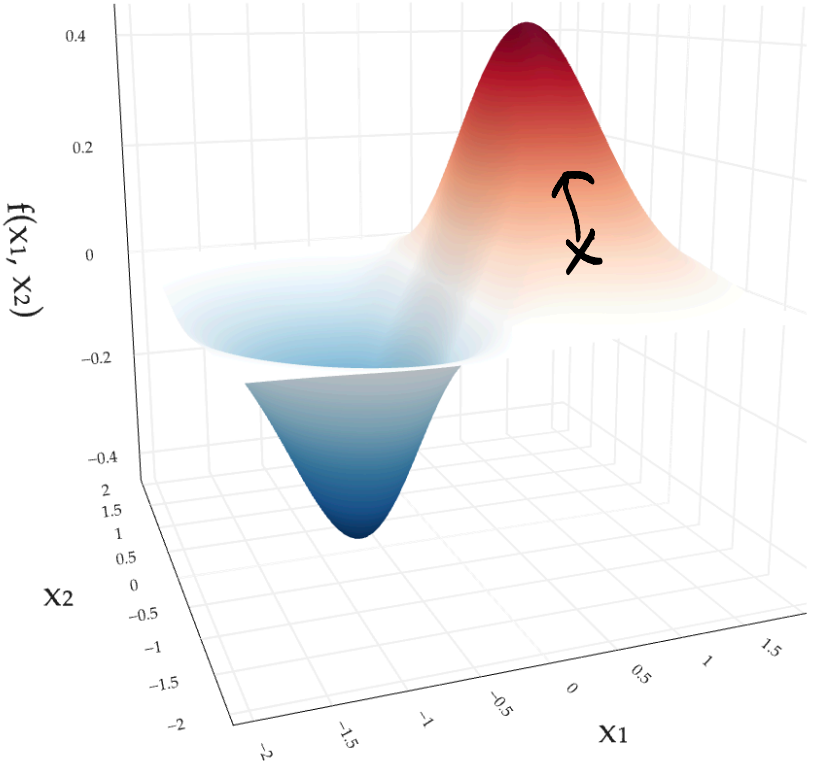
Agenda

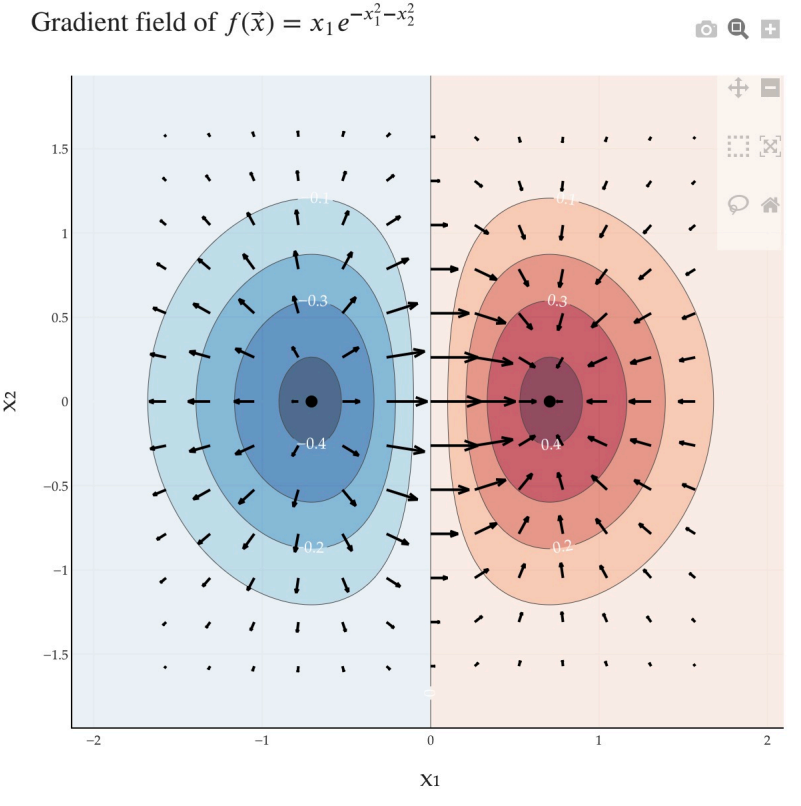
- Recap: gradients } 8.1
- "Big 3" gradient rules } 8.2
- Minimizing mean squared error using gradients
- How do we minimize functions where $\nabla f(\vec{w})$ exists but solving $\nabla f(\vec{w}) = \vec{0}$ is hard? ⇒ gradient descent } 8.3

Announcements

- HW 7 due Sunday
- HW 6 + yesterday's lab solutions posted (including a video for the chicken/beef/fish Q!)

$f: \mathbb{R}^d \to \mathbb{R}$ "vector-to-scalar" "contour plot"





Recall, if $f: \mathbb{R}^d \to \mathbb{R}$,

f: vector-to-scalar

$$\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

∇f: vector-to-vector

e.g.

$$f(\vec{x}) = (x_1 + x_2)^2 + 2\cos(x_1)$$

$$\nabla f(\vec{x}) = \begin{bmatrix} 2(x_1 + x_2) - 2\sin(x_1) \\ 2(x_1 + x_2) \end{bmatrix}$$

Ultimate goal: find gradient of

multiple linear regression

$$R_{sq}(\vec{w}) = \frac{1}{n} \|\vec{y} - X\vec{w}\|^2$$

$$R_{sq} : \mathbb{R}^{d+1} \to \mathbb{R}$$

Big 3 gradient rules

① $f(\vec{x}) = \vec{a} \cdot \vec{x} = \vec{a}^T \vec{x}$ $\quad \nabla f(\vec{x}) = \vec{a}$

② $f(\vec{x}) = \|\vec{x}\|^2$ $\quad \nabla f(\vec{x}) = 2\vec{x}$

③ $f(\vec{x}) = \vec{x}^T A \vec{x}$ $\quad \nabla f(\vec{x}) = (A + A^T)\vec{x}$

"quadratic form"

if A symmetric:

$\nabla f(\vec{x}) = 2A\vec{x}$

$$f(\vec{x}) = \|\vec{x}\|$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \qquad f(\vec{x}) = \sqrt{x_1^2 + x_2^2 + \cdots + x_d^2}$$

could find $\frac{\partial f}{\partial x_1} = \frac{1}{2\sqrt{x_1^2 + x_2^2 + \cdots + x_d^2}} \cdot 2x_1 = \frac{x_1}{\|\vec{x}\|}$

similarly, $\frac{\partial f}{\partial x_2} = \frac{x_2}{\|\vec{x}\|}, \cdots$ $\nabla f(\vec{x}) = \begin{bmatrix} x_1/\|\vec{x}\| \\ x_2/\|\vec{x}\| \\ \vdots \\ x_d/\|\vec{x}\| \end{bmatrix} = \frac{\vec{x}}{\|\vec{x}\|}$

$$f(\vec{x}) = \|\vec{x}\|$$

$$\vec{x} \to \|\vec{x}\|^2 \to \|\vec{x}\|$$

g: vector to scalar

h: scalar to scalar

Ideally, we use the fact that $\nabla \|\vec{x}\|^2 = 2\vec{x}$

$$g(\vec{x}) = \|\vec{x}\|^2$$

$$h(y) = \sqrt{y}$$

$$h'(y) = \frac{1}{2\sqrt{y}}$$

$$f(\vec{x}) = h(g(\vec{x}))$$

chain rule:

$$\nabla f(\vec{x}) = h'(g(\vec{x}))\, \nabla g(\vec{x})$$

Here: $\nabla \|\vec{x}\| = \frac{1}{2\sqrt{g(\vec{x})}} \nabla g(\vec{x}) = \frac{1}{2\|\vec{x}\|}(2\vec{x}) = \frac{\vec{x}}{\|\vec{x}\|}$

$$f(\vec{x}) = (\vec{a}\cdot\vec{x})^2$$

① without chain rule

$$f(\vec{x}) = (\vec{a}\cdot\vec{x})^2 = (\vec{x}^T\vec{a})(\vec{a}^T\vec{x})$$

$$= \vec{x}^T(\vec{a}\vec{a}^T)\vec{x}$$

quadratic form

$\vec{a}\vec{a}^T$ symmetric matrix!

$$\nabla f(\vec{x}) = 2\vec{a}\vec{a}^T\vec{x}$$

$$= 2\vec{a}(\vec{a}^T\vec{x})$$

$\vec{a}^T\vec{x}$ → scalar!

$$= 2(\vec{a}\cdot\vec{x})\vec{a}$$

② with chain rule:

$$\nabla(\vec{a}\cdot\vec{x})^2 = 2(\vec{a}\cdot\vec{x})\nabla(\vec{a}\cdot\vec{x})$$

$$= 2(\vec{a}\cdot\vec{x})\vec{a}$$

$$R_{sq}(\vec{w}) = \frac{1}{n}\|\vec{y} - X\vec{w}\|^2$$

$$= \frac{1}{n}(\vec{y} - X\vec{w})^T(\vec{y} - X\vec{w})$$

$$= \frac{1}{n}\left(\vec{y}^T\vec{y} - \vec{y}^T X\vec{w} - (X\vec{w})^T\vec{y} + (X\vec{w})^T X\vec{w}\right)$$

$$= \frac{1}{n}\left(\vec{y}^T\vec{y} - 2\vec{w}^T(X^T\vec{y}) + \vec{w}^T(X^T X)\vec{w}\right)$$

symmetric!

$$\nabla R_{sq}(\vec{w}) = \frac{1}{n}\left(-2X^T\vec{y} + 2X^T X\vec{w}\right)$$

$$\vec{y}^T X \vec{w} \quad = \quad (X\vec{w})^T \vec{y}$$

both are dot product of $\vec{y}$ with $X\vec{w}$!

$$\vec{y}^T X \vec{w} = (X\vec{w})^T \vec{y} = \vec{w}^T (X^T \vec{y}) = \vec{w} \cdot (X^T \vec{y})$$

$$\nabla R_{sq}(\vec{w}) = \frac{1}{n}\left(-2X^T\vec{y} + 2X^TX\vec{w}\right)$$

$$= -\frac{2}{n}\left(X^T\vec{y} - X^TX\vec{w}\right)$$

$$\nabla R_{sq}(\vec{w}) = \vec{0}$$

$$X^T\vec{y} - X^TX\vec{w} = \vec{0}$$

$$X^TX\vec{w} = X^T\vec{y}$$

normal equation from before!

if X^TX invertible,
i.e. if X's cols are linearly independent

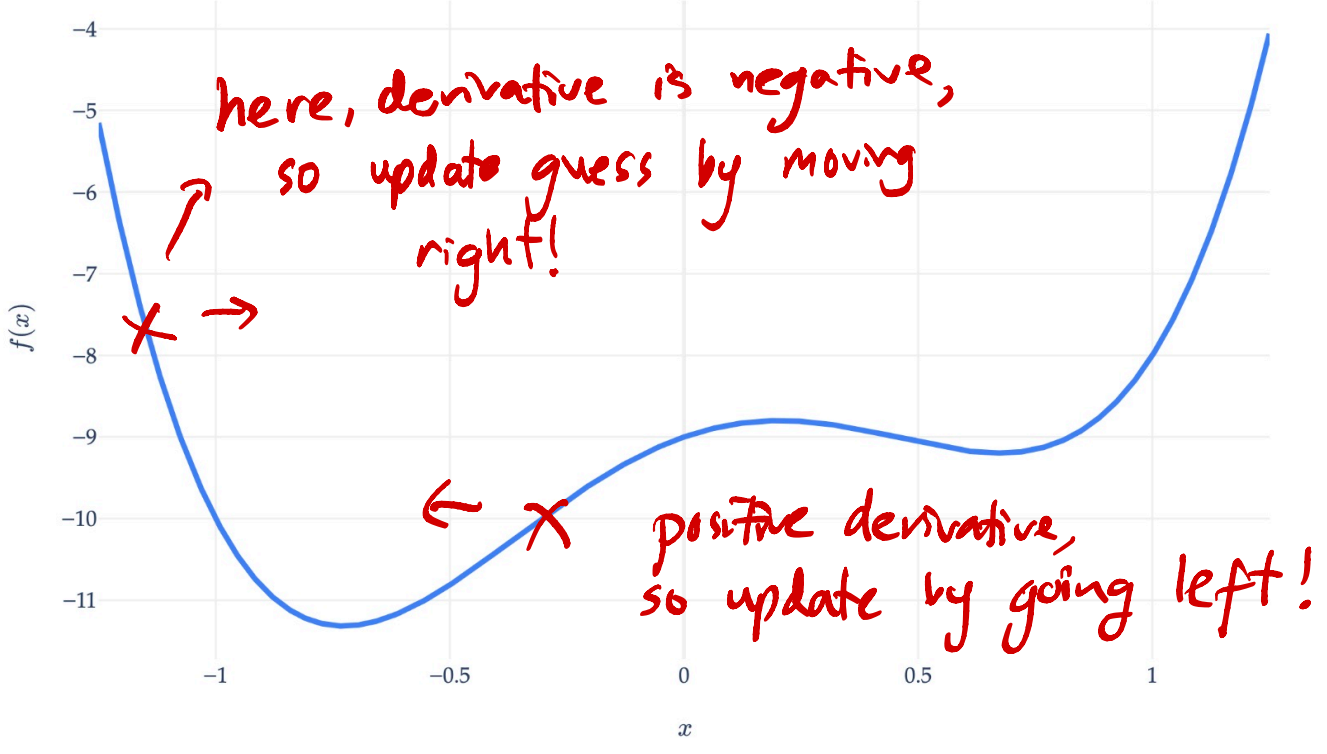
$$\vec{w}^* = (X^TX)^{-1}X^T\vec{y}$$

now: suppose we've calculated $\nabla f(\vec{x})$, but it is hard/impossible to solve

$$\nabla f(\vec{x}) = \vec{0}.$$

then what?

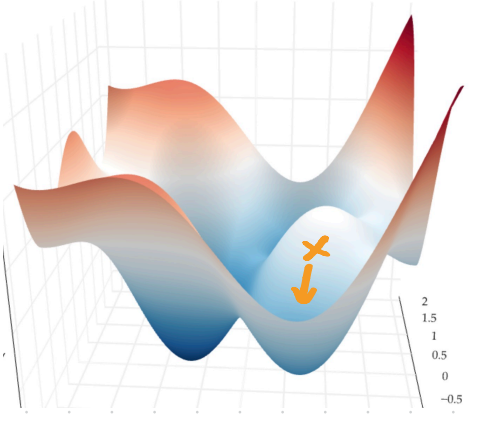
$$f(x) = 5x^4 - x^3 - 5x^2 + 2x - 9$$



Gradient descent

Goal: minimize $f : \mathbb{R}^d \to \mathbb{R}$, given $\nabla f(\vec{x})$

- First, choose an initial guess, $\vec{x}^{(0)}$, and learning rate / step size, $\alpha > 0$

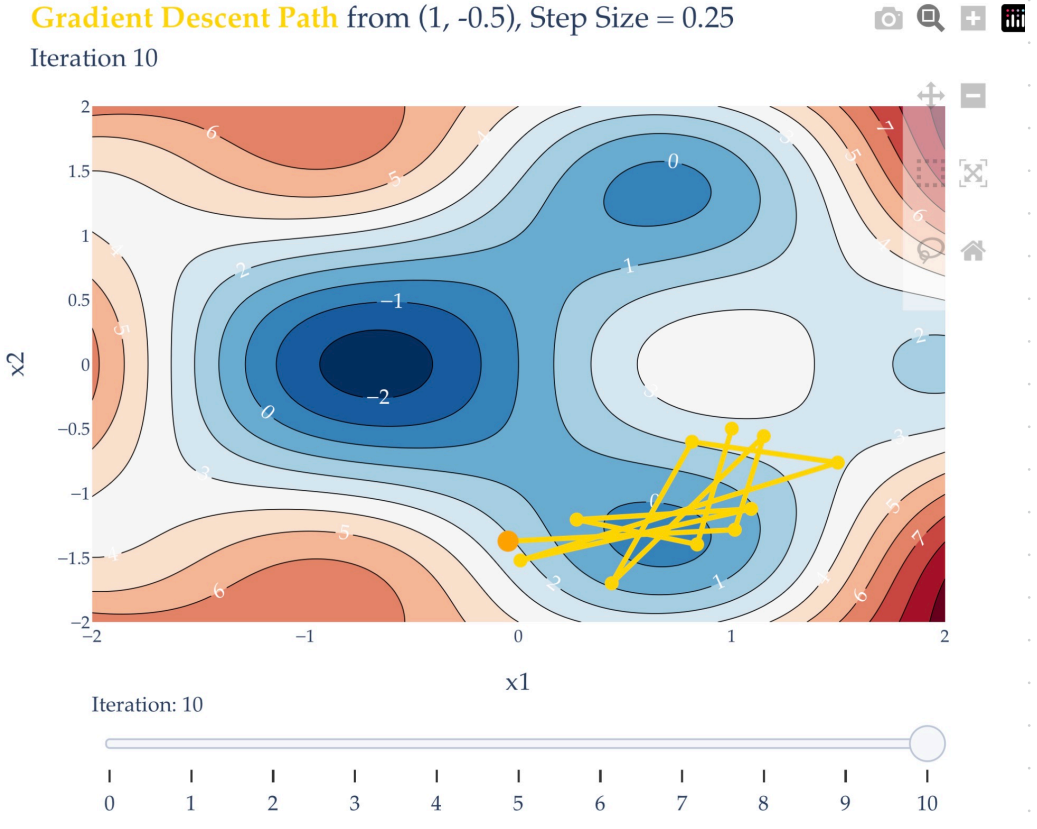


- Update guess iteratively using:

$$\vec{x}^{(t+1)} = \vec{x}^{(t)} - \alpha \nabla f(\vec{x}^{(t)})$$

move against the gradient

- Terminate when $\| \nabla f(\vec{x}^{(t)}) \| \leq 0.0001$



lots of examples in Ch. 8.3 :

play around with it